System architecture

How the pieces fit together — existing tools doing what they already do well, with just enough custom glue to stop duplicates, dropped handoffs, and silent failures.

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Architecture principle

The rule is simple: let each existing tool do what it's already good at, and build only the glue that prevents duplicates, missing next steps, bad handoffs, and invisible failures. Nothing gets rebuilt that already works.

Operating system map

Every lead source funnels through GHL, hands off to JobNimbus as the job of record, and fans out to documents, materials, finance, and the owner's exceptions view. GHL and JobNimbus carry the signal — everything else mirrors or archives.

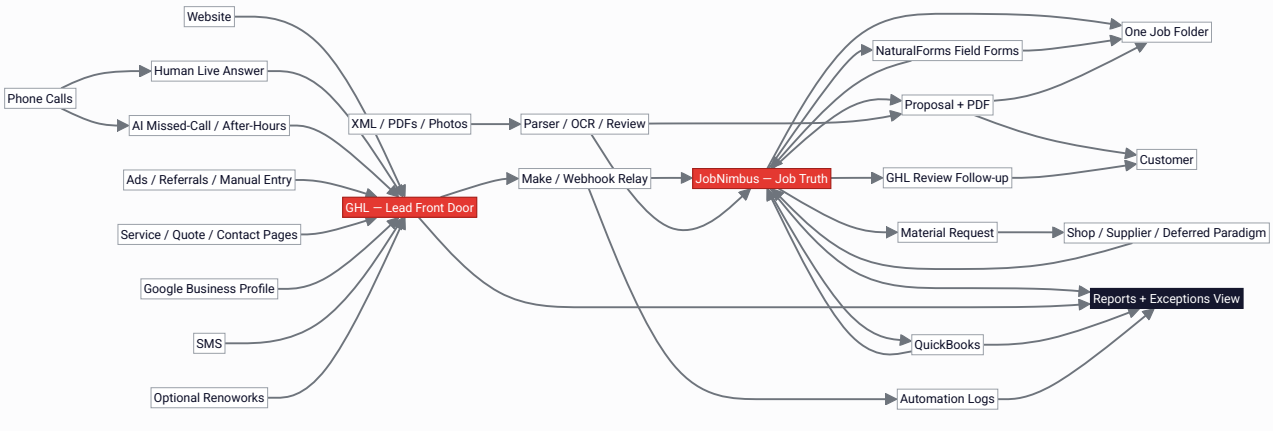


FIGURE 1 — LEAD-TO-CASH OPERATING MAP

Website & front-door funnel

Every inbound channel enters GHL first — the home page form, the quote and contact forms, the service-page forms with hidden service fields, the city / neighborhood and blog / comparison CTAs, Google Business Profile, phone, SMS, and referrals. GHL captures source URL, landing page, page type, service intent, location, UTM and call source, attachment, and message, then hands qualified leads to JobNimbus. Make does the dedupe, idempotent folder links, and failure alerts on the way through. Renoworks stays deferred unless visualizer-led selling is approved.

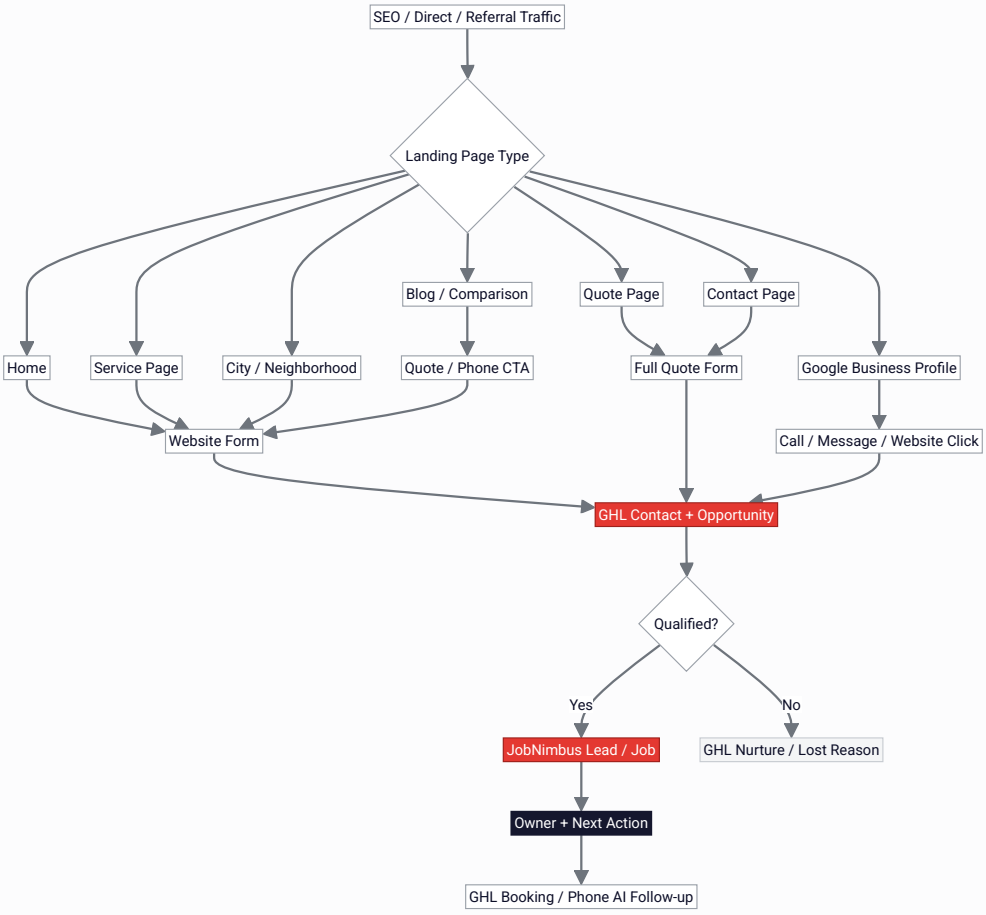


FIGURE 2 — WEBSITE & FRONT-DOOR FUNNEL INTO GHL, THEN JOBNIMBUS

First-version flow

The first build only has to move a lead cleanly from intake to a job, with dedupe in the middle and a folder created once the canonical job number exists.

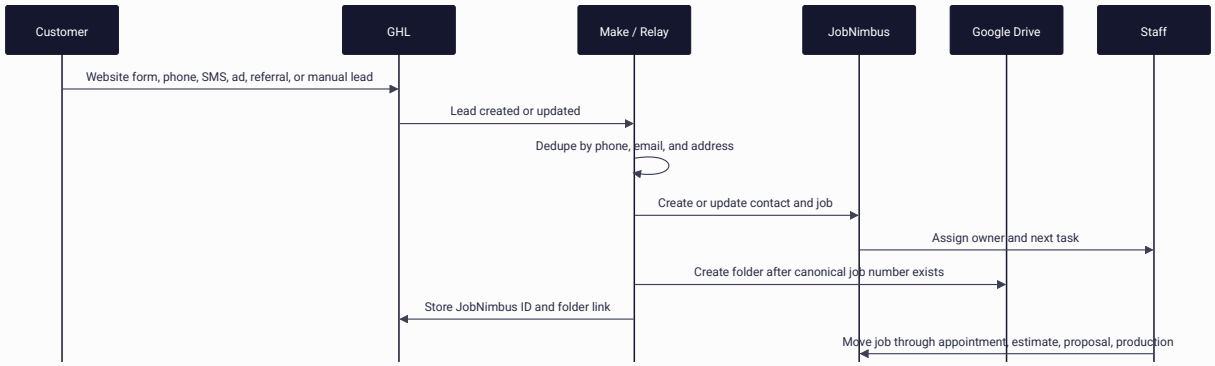


FIGURE 3 — FIRST-VERSION LEAD SEQUENCE, INTAKE TO JOB

End-state flow

Where it lands once every layer is in place — sales, finance, field, and materials all reporting status up into a single owner exceptions view.

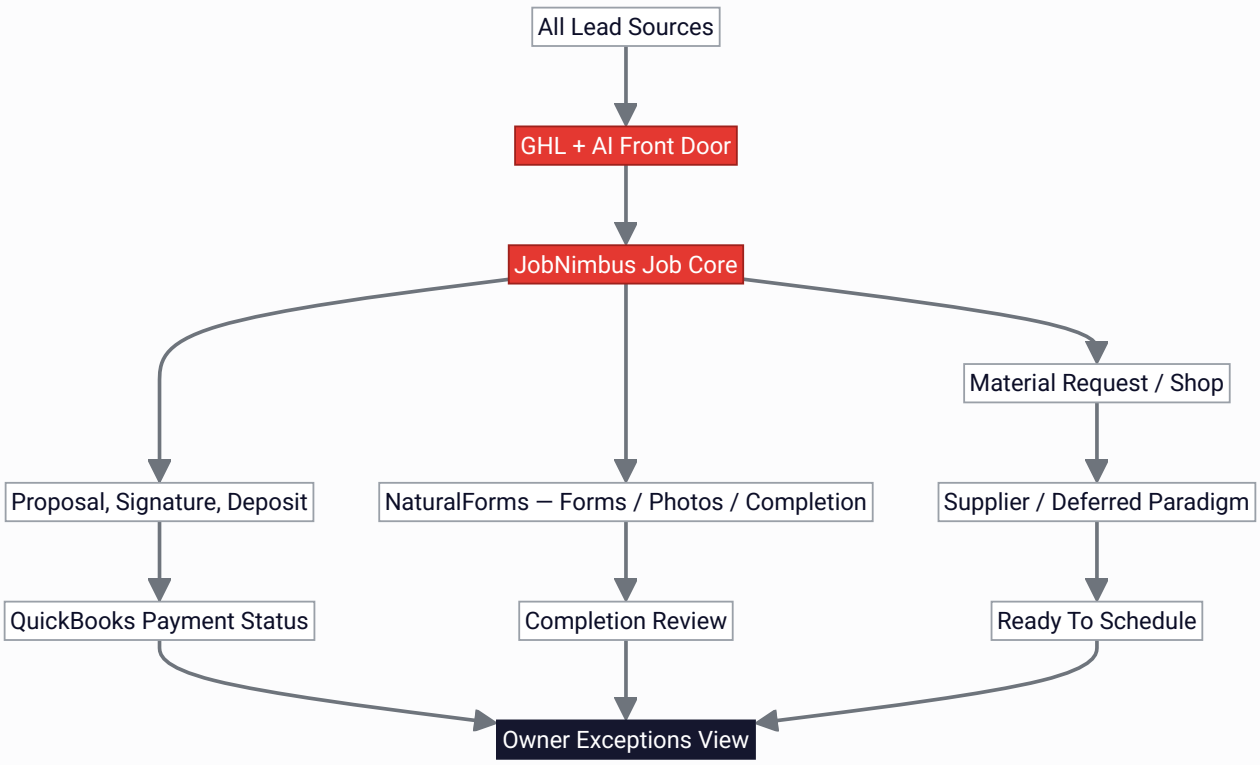


FIGURE 4 — END-STATE FLOW ACROSS EVERY DIVISION

Component responsibilities

Each system has a tight remit — and an explicit list of what it must *not* try to own.

COMPONENT	OWNS	DOES NOT OWN
GHL	Lead inbox, conversations, SMS, booking, review requests, AI pilot	Canonical job number, accounting, final job truth
JobNimbus	Customer / job / property / stage / task / owner truth	Accounting ledger, broad marketing nurture
NaturalForms	Field forms, signatures, photos, sketches, completed-form evidence	Canonical job stage, accounting, final pricing
Make	Workflow glue, app triggers, routing, transport	System-of-record decisions, unlogged critical mutations
Webhook relay (if built)	Dedupe, idempotency, logging, validation	CRM replacement
Google Drive	Job archive and attachments	Source of truth for job stage or payment status
QuickBooks	Invoices, payments, accounting records	CRM pipeline, job operations
Parser / OCR	Extracts document data and a confidence score	Final pricing without human review
Proposal generator	Customer-facing proposal and PDF archive	Final accounting invoice

Minimal custom components

Build these only when an existing tool genuinely fails the requirement — not before:

1. Dedupe and idempotency service.
2. Webhook relay and audit logger — if Make's own logs and errors aren't enough.
3. AppliCad XML parser.
4. PDF / OCR extraction queue.
5. Proposal HTML / PDF generator.
6. AI receptionist tool handlers, once a provider is chosen.
7. Lightweight exceptions view — if reports and a sheet fall short.
8. Integration health checker.

Integration contract

Every integration that mutates records has to declare all of this up front. No mutation ships without it.

- Owner.
- Trigger.
- Input fields.
- Output fields.
- Idempotency key.
- Retry behavior.
- Failure alert.
- Manual recovery step.
- Kill switch.
- Test record.